Rahul Anand Sharma

ML Systems Engineer & Researcher · LLM Serving Architecture · Accelerator Scaling (TPU/GPU) · Hardware-Software Co-Design

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Executive Summary

ML Systems Engineer and Researcher specializing in LLM serving architecture, compiler-hardware co-design, and distributed accelerator systems (TPUs/GPUs). Ph.D. in Electrical & Computer Engineering from Carnegie Mellon University with 15+ publications at premier systems and security venues (SIGCOMM, NSDI, USENIX Security, CoNEXT) and 2 granted US patents. Proven track record of architecting and scaling frontier AI infrastructure across **Google Cloud** (Gemini inference optimization, TPU v7 Ironwood multi-cube scaling, Sovereign Cloud disaggregated serving), **AWS** (Amazon Q Spark auto-debugging at hyperscale), and **Microsoft Research**.

Professional Experience

Google Cloud
Aug. 2025 – Present
ML Systems Engineer — *Mountain View, CA*

- **Gemini Serving & Runtime Optimization:** Driving cross-stack inference performance for Gemini models across production GPU and TPU fleets, targeting XLA/runtime bottleneck elimination, KV-cache memory efficiency, and end-to-end TTFT/TPOT latency.
- **Sovereign Cloud Disaggregated Serving (GDC):** Spearheaded the intra-node (4:4 prefill/decode split) disaggregated serving architecture for Gemini on bare-metal nodes in Google Distributed Cloud, overcoming strict air-gapped memory isolation and control-plane constraints.
- **TPU v7 (Ironwood) Architecture & Multi-Cube Scaling:** Led the TPU v7 NPI microbenchmarking suite bridging internal Borg cluster infrastructure with Cloud XLA/JAX runtimes; executed multi-cube scaling analyses to model ICI/DCN interconnect limits across large-scale topologies.
- **Frontier AI Customer Enablement:** Delivered collective communication (`AllReduce/AllGather/AllToAll`) microbenchmarks and public scaling recipes for the TPU v7 preview, establishing performance baselines that anchored multi-billion-dollar hardware evaluations with tier-1 frontier AI labs.
- **Day-1 3P Model Sim-Ship (*Google Cloud Tech Impact Award, 2026*):** Architected automated continuous regression pipelines enforcing Borg–Cloud runtime parity and led XLA/kernel optimizations for open-weight frontier LLMs (Llama 3, DeepSeek) to land Day-1 hardware enablement.

AWS, Amazon
Jan. 2023 – July 2025
Applied Scientist — *Palo Alto, CA*

- **Hyperscale Spark Auto-Debugging (Amazon Q):** Architected and launched a Generative AI root-cause analysis system for Apache Spark across AWS Glue, Amazon EMR, and internal compute fleets, leveraging LLMs over distributed execution logs and DAG telemetry to slash customer Mean Time to Resolution (MTTR) from days to minutes.
- **Active Cloud Network Security Verification:** Engineered a high-throughput active testing service for AWS network security, injecting synthetic live traffic to validate Network ACLs at scale, guaranteeing strict compliance and eliminating false-positive blindspots inherent to static simulation models.
- **Production Control-Plane Diagnostics:** Designed and shipped real-time failure classification and automated remediation pipelines directly into AWS Glue and EMR management consoles, serving thousands of enterprise analytics customers.

Microsoft Research
May 2020 – July 2021
Research Intern, Networking Research Group — *Redmond, WA*

- **Edge-Cloud DNN Offloading Protocols:** Architected low-latency distributed transport and adaptive partitioning protocols for edge-cloud deep learning inference offloading, minimizing tail latency and communication overhead under bandwidth-constrained WAN links.

Microsoft Research
July 2016 – July 2018
Research Fellow, Mobility, Networks & Systems Group — *Bangalore, India*

- **Edge-AI Workload Scheduling (DeepEdge):** Pioneered a distributed edge-AI scheduling architecture optimizing multi-tenant deep learning pipelines under severe compute and bandwidth constraints with graceful degradation (*published at USENIX NSDI '19; granted US Patent 10,942,767*).
- **FarmBeats IoT & Cloud Platform (*Microsoft Outstanding Technical Achievement Award*):** Co-designed and built an end-to-end RF/sensor and cloud analytics platform for data-driven agriculture; productized into Microsoft Azure and featured by Bill Gates (*granted US Patent 11,327,476*).
- **Neural Network Learnability & Mentorship:** Investigated empirical generalization bounds governing network depth and minibatch dynamics with theoretical ML researchers; mentored 4+ research interns across systems projects resulting in multiple top-tier publications.

Education

Carnegie Mellon University
Aug. 2018 – Jan. 2023
Ph.D. in Electrical & Computer Engineering (CyLab Security & Privacy Institute) — *Pittsburgh, PA*

- **Thesis:** *Practical Network Layer Machine Learning for IoT Security* | **Advisors:** Prof. Vyas Sekar & Prof. Anthony Rowe | **Honors:** CIT Dean's Fellowship

International Institute of Information Technology (IIIT)
July 2011 – July 2016
B.Tech and M.S. by Research in Computer Science — *Hyderabad, India*

- **Honors:** Dean's List for Academic and Research Excellence (3 consecutive years) | **Thesis:** *Automated Broadcast Sports Video Analytics*

Technical Expertise

LLM Serving & Distributed ML:	Disaggregated Serving (Prefill/Decode), KV-Cache Optimization, Model Parallelism (TP/PP/EP), Collective Communication (`AllReduce/AllGather/AllToAll`), vLLM, FSDP, Megatron-LM, Speculative Decoding, Quantization (FP8/INT4)
Compilers, Frameworks & Kernels:	JAX, XLA / HLO, PyTorch, Triton, TensorRT / TensorRT-LLM, CUDA, C++, Python, SQL, Bash
Accelerators & Cloud Systems:	Google Cloud TPU (v4/v5p/v7 Ironwood), NVIDIA GPU Clusters, Google Borg, Google Distributed Cloud (GDC / Air-Gapped Sovereign Cloud), AWS (Glue, EMR, VPC), Apache Spark, Kubernetes, Ray

Granted US Patents & Honors

- **US Patent 10,942,767 (Granted 2021):** *Deep Neural Network Workload Scheduling* — Distributed scheduling architecture for optimizing deep learning inference workloads across heterogeneous edge and cloud compute under strict resource constraints.
- **US Patent 11,327,476 / US20200150640A1 (Granted 2022):** *Sensor Fall-Curve Identification* — Time-of-flight and RF signal fall-curve analytics for fault isolation in large-scale IoT deployments (productized in Microsoft Azure FarmBeats).
- **Google Cloud Tech Impact Award (2026):** Recognized for TPU v7 (Ironwood) collective communication scaling baselines, Borg–Cloud runtime parity pipelines, and Day-1 frontier LLM enablement (Sim-Ship).
- **Microsoft Outstanding Technical Achievement Award (2018):** Awarded for co-designing and engineering FarmBeats, transitioning research into a commercial Microsoft Azure product.
- **Academic Fellowships:** CIT Dean's Doctoral Fellowship, Carnegie Mellon University (2018); Dean's List for Academic and Research Excellence, IIIT Hyderabad (2014–2016).
- **Executive & Media Recognition:** Research featured by **Bill Gates** (*GatesNotes*), **Microsoft CEO Satya Nadella** (Keynote Address), *Der Spiegel*, *The Washington Post*, *The Register*, and *The Hacker News*.

Peer-Reviewed Publications (450+ Citations)

Systems, Networking, Security & Distributed ML

1. **[ACM CoNEXT '22] R. A. Sharma**, I. Sabane, M. Apostolaki, A. Rowe, and V. Sekar. "Lumen: A Framework for Developing and Evaluating ML-Based IoT Network Anomaly Detection." *Proc. of ACM CoNEXT*, Rome, Italy, Dec. 2022.
   Impact: Modular evaluation framework across 100+ devices isolating temporal and topological domain-shift degradation in network anomaly detection.
2. **[USENIX Security '22] R. A. Sharma**, E. Soltanaghaei, A. Rowe, and V. Sekar. "Lumos: Identifying and Localizing Diverse Hidden IoT Devices in an Unfamiliar Environment." *Proc. of the 31st USENIX Security Symposium*, Boston, MA, Aug. 2022.
   Impact: Mobile wireless sensing system combining 802.11 PHY/MAC fingerprinting and AR to localize hidden IoT devices to 1.5m accuracy (95% precision).
3. **[USENIX Security '21]** S.-J. Moon, Y. Yin, **R. A. Sharma**, Y. Yuan, J. M. Spring, and V. Sekar. "Accurately Measuring Global Risk of Amplification Attacks using AmpMap." *Proc. of the 30th USENIX Security Symposium*, Aug. 2021.
   Impact: Internet-wide active measurement and ML inference pipeline quantifying amplification DDoS attack surfaces across millions of endpoints.
4. **[ACM SIGCOMM '20]** A. Manousis, **R. A. Sharma**, V. Sekar, and J. Sherry. "Contention-Aware Performance Prediction for Virtualized Network Functions." *Proc. of ACM SIGCOMM*, Aug. 2020.
   Impact: Low-level hardware contention modeling (LLC, memory bus, packet I/O) to predict throughput degradation and guarantee QoS SLOs in multi-tenant cloud NFV.
5. **[ACM/IEEE IPSN '20] R. A. Sharma**, A. Dongare, J. Miller, N. Wilkerson, D. Cohen, V. Sekar, P. Dutta, and A. Rowe. "All that GLITTERs: Low-Power Spoof-Resilient Optical Markers for Augmented Reality." *Proc. of ACM/IEEE IPSN*, Sydney, Australia, Apr. 2020.
   Impact: BLE-synchronized optical sensing pipeline enabling 3D pose estimation and spoof-resilient AR localization at 30m range and milliwatt power.
6. **[ACM BuildSys '20]** E. Soltanaghaei, **R. A. Sharma**, Z. Wang, A. Chittilappilly, A. Luong, E. Giler, K. Hall, S. Elias, and A. Rowe. "Robust and Practical WiFi Human Sensing Using On-Device Learning with a Domain Adaptive Model." *Proc. of ACM BuildSys*, Nov. 2020.
7. **[USENIX NSDI '19] R. A. Sharma** and R. Chandra. "DeepEdge: A Network Edge for Deep Learning Workloads." *16th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, Boston, MA, Feb. 2019.
   Impact: Distributed edge-cloud DNN inference scheduler optimizing multi-stream video analytics under severe bandwidth constraints (US Patent 10,942,767).
8. **[ACM COMPASS '19]** A. N. Swamy, A. Kumar, R. Patil, A. Jain, Z. Kapetanovic, **R. A. Sharma**, D. Vasisht, M. Swaminathan, R. Chandra, and A. Badam. "Low-Cost Aerial Imaging for Small Holder Farmers." *Proc. of ACM COMPASS*, Accra, Ghana, July 2019.
9. **[ACM SenSys '18]** T. Chakraborty, A. U. Nambi, R. Chandra, **R. A. Sharma**, M. Swaminathan, Z. Kapetanovic, and J. Appavoo. "Fall-Curve: A Novel Primitive for IoT Fault Detection and Isolation." *Proc. of ACM SenSys*, Shenzhen, China, Nov. 2018. *(US Patent 11,327,476)*
10. **[MSR Tech Report '18] R. A. Sharma**, N. Goyal, M. Choudhury, and P. Netrapalli. "Learnability of Learned Neural Networks." *arXiv preprint arXiv:1806.00997 / Microsoft Research*, 2018.

Computer Vision & Video Understanding (Early Research)

11. **[IEEE WACV '18] R. A. Sharma**, B. Bhat, V. Gandhi, and C. V. Jawahar. "Automated Top View Registration of Broadcast Football Videos." *IEEE Winter Conference on Applications of Computer Vision (WACV)*, Lake Tahoe, NV, Mar. 2018.
12. **[Springer SIVP '17] R. A. Sharma**, V. Gandhi, V. Chari, and C. V. Jawahar. "Automatic Analysis of Broadcast Football Videos Using Contextual Priors." *Signal, Image and Video Processing*, Vol. 11, No. 1, pp. 171–178, 2017.
13. **[ACM ICVGIP '16]** H. Saraogi, **R. A. Sharma**, and V. Kumar. "Event Recognition in Broadcast Soccer Videos." *Proc. of the 10th Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, Dec. 2016.
14. **[IEEE ACPR '15] R. A. Sharma**, K. P. Sankar, and C. V. Jawahar. "Fine-Grain Annotation of Cricket Videos." *Proc. of the 3rd IAPR Asian Conference on Pattern Recognition (ACPR)*, Kuala Lumpur, Malaysia, Nov. 2015. *(Featured in Satya Nadella Keynote & The Washington Post)*

Selected Invited Talks & Presentations

- **USENIX Security Symposium (2022):** "Lumos: Identifying and Localizing Diverse Hidden IoT Devices" — *Boston, MA*
- **CMU CyLab & CONIX Annual Review (2022):** "Lumen: ML-Based IoT Network Anomaly Detection" — *Pittsburgh, PA*
- **ACM/IEEE IPSN & CPS-IoT Week (2020):** "All that GLITTERs: Light Anchors for Augmented Reality" — *Sydney, Australia*
- **Microsoft Research TechFest (2018):** "FarmBeats: AI, Edge & IoT for Data-Driven Agriculture" — *Redmond, WA*
- **IEEE WACV (2018):** "Automated Top-View Registration of Broadcast Sports Video" — *Lake Tahoe, NV*